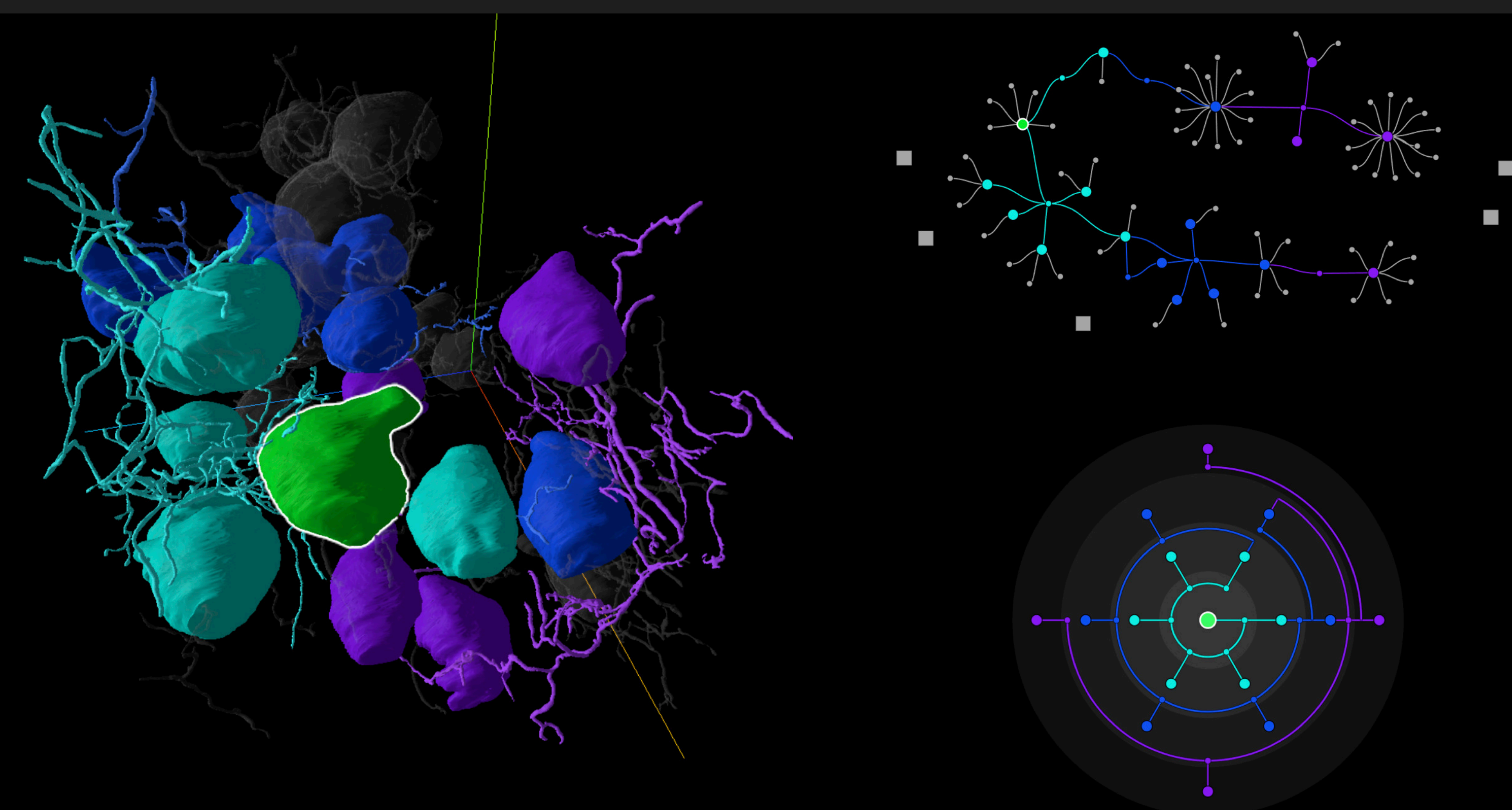
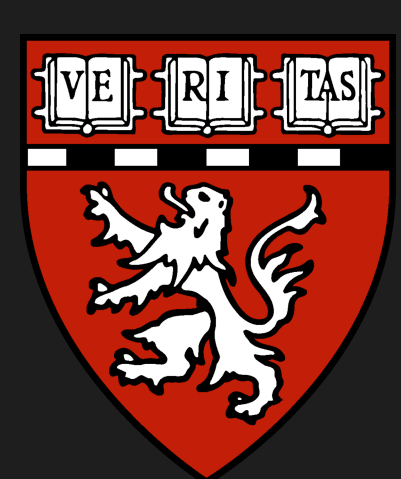


Nexus: Interactive Visual Exploration of 3D Spatial Connectivity for Tissue Analysis

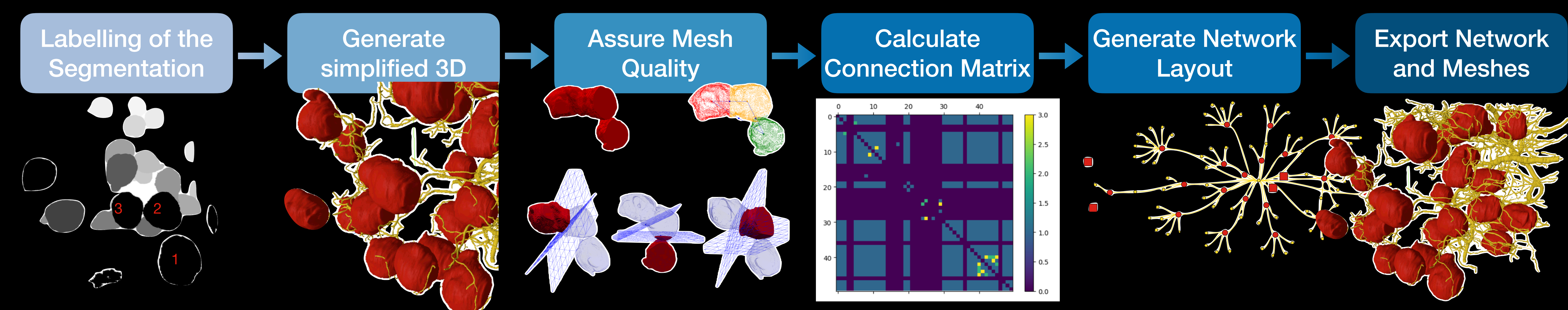
Eric Mörth, Benjamin Beinder, Renata G. Raidou, Nils Gehlenborg



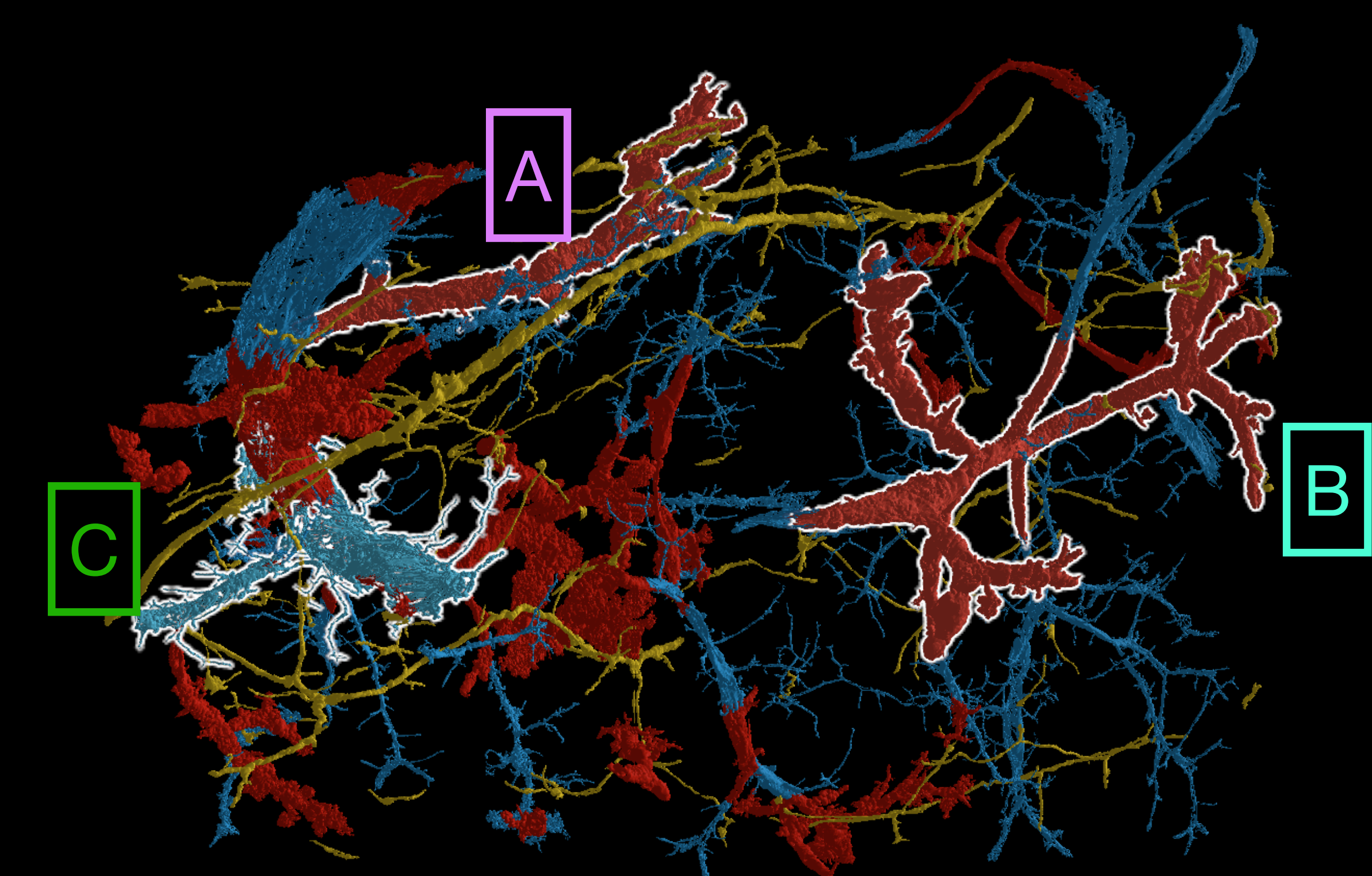
Nexus: Network-based EXploration and Connectivity Analysis for Understanding Spatial Tissue Organization - unifies spatial and network views to analyze connectivity in high-resolution 3D tissue imaging.

Nexus reveals multiscale tissue connectivity, linking 3D spatial context with network views. It advances insight into organ function, disease progression, and cellular interactions.

Figure 1. Nexus integrates spatial and network views for tissue connectivity analysis.



Case Study Pancreas:



From Segmentations to Insights

- Nexus transforms raw segmentations into meshes and connectivity networks
- An integrated visualization platform enables expert exploration and discovery
- Three case studies demonstrate its utility across diverse tissues

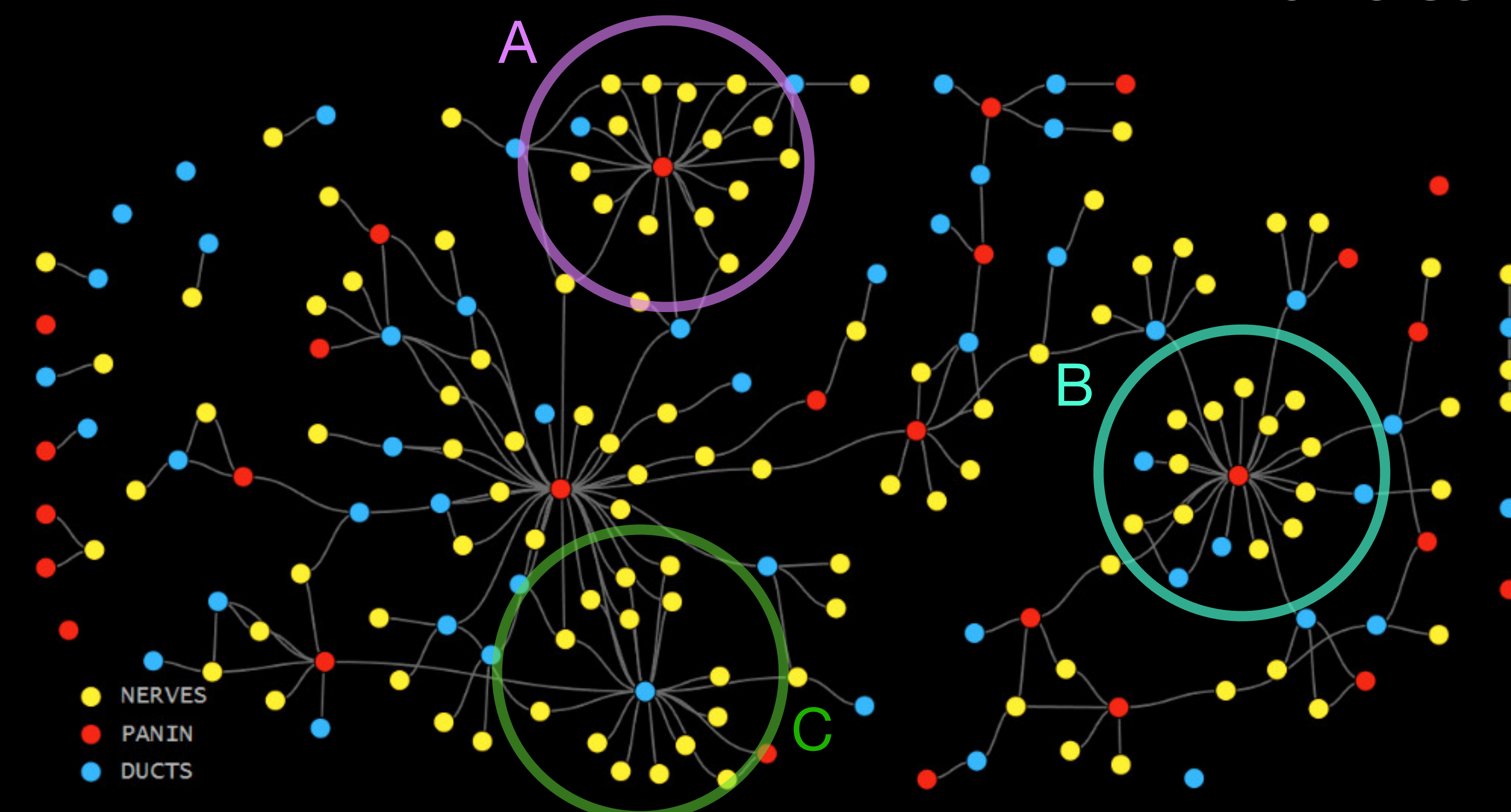


Figure 2. Nexus applied to pancreas data (Kiemen et al.), highlighting differences in neural innervation (yellow) between PanINs (red) and normal ductal tissue (blue).

Case Study Melanoma:

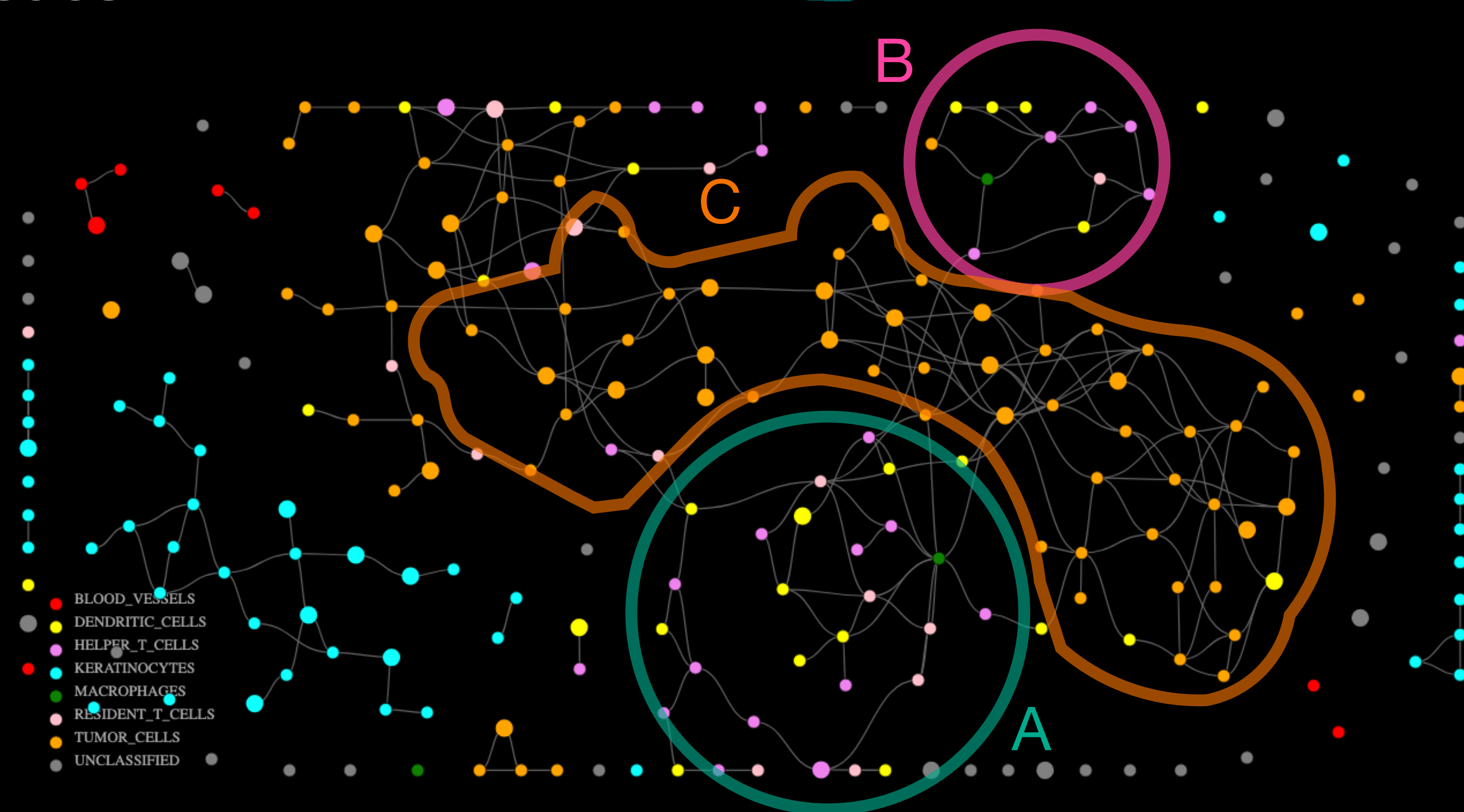
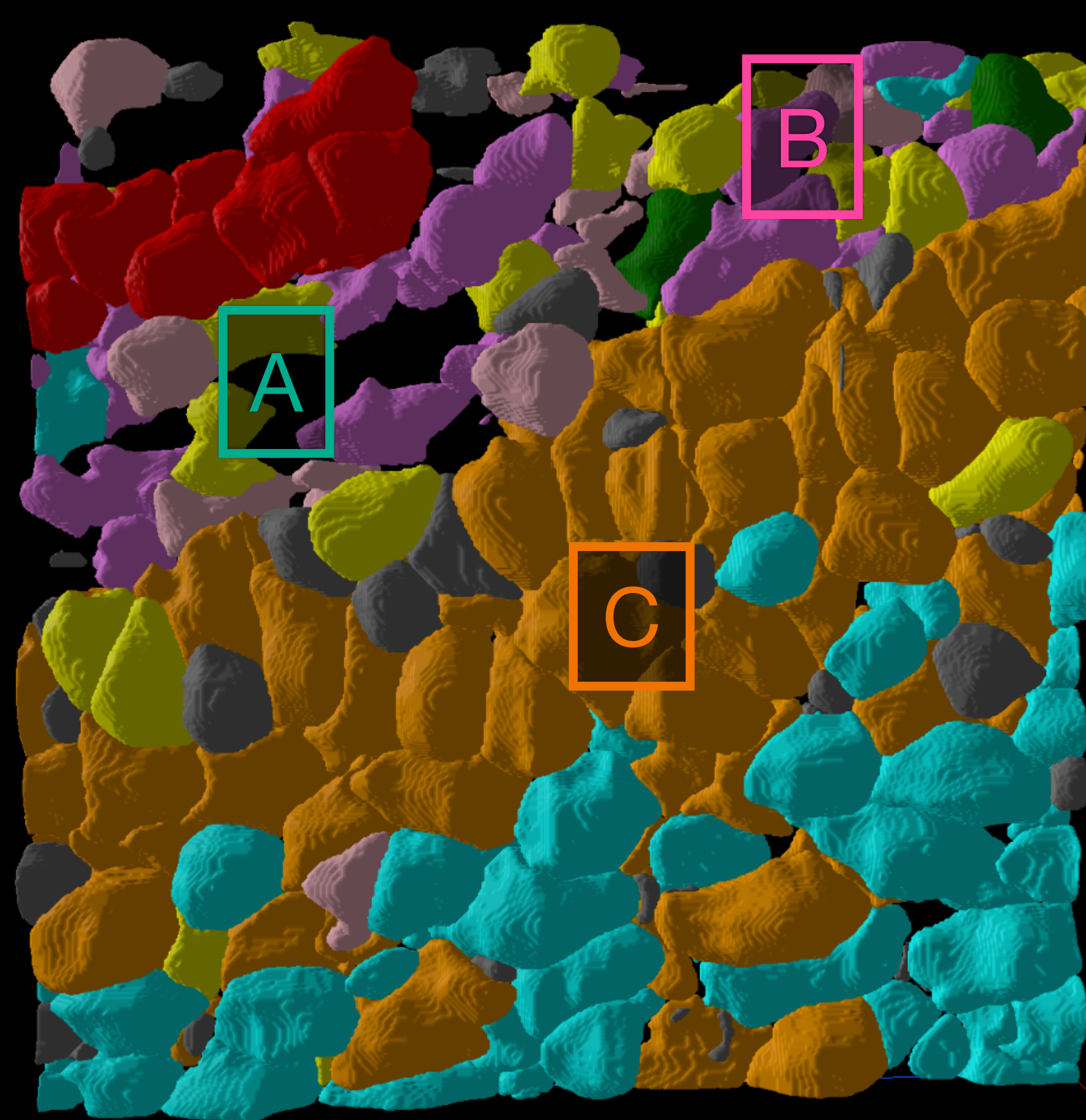


Figure 3. Nexus applied to skin cancer dataset (Yapp et al.), tumor (orange) and immune boundary (green and pink) is clearly visible in the network abstraction.

Future Work: Integrate into Viteessce Support Scalability Explore more tissue types